

- 13.** Consider the following statements:
 I. The solution of equation $2x + 3y = 425$ and $3x + 2y = 350$ is a positive integral pair.
 II. for $k = -2$, the equations $x - ky = 2$ and $3x + 6y = -5$ have infinitely many solutions.
 III. $(2, 5)$ and $(-1, 3)$ are solutions of $2x + 5y = 13$
 Select the correct answer using the codes given below:
 (a) Only I (b) Both I and II
 (c) Both II and III (d) All of these
- 14.** A horse and two cows together cost ₹ 680. If a horse costs ₹ 80 more than a cow, then the cost of horse is
 (a) ₹ 170 (b) ₹ 280 (c) ₹ 200 (d) ₹ 220
- 15.** Sunita has 10 paise and 50 paise coins in her purse. If the total number of coins is 17 and their total value is ₹ 4.50, then number of 10 paise coins is
 (a) 9 (b) 7 (c) 10 (d) 5
- 16.** The solution of the equation $\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$ is given by which one of the following?
 (a) $x = 2, y = 1$ (b) $x = 1, y = 1$
 (c) $x = -1, y = -1$ (d) $x = 1, y = 2$
- 17.** The line $3x - 5y = -10$ cuts Y-axis at
 (a) $(0, 2)$ (b) $(0, 1)$ (c) $(0, 3)$ (d) $(0, 4)$
- 18.** The equations $px + q = 0$ and $rx + s = 0$ are consistent, if
 (a) $ps = qr$ (b) $ps + qr = 0$ (c) $pq - rs = 0$ (d) $pq + rs = 0$
- 19.** A streamer goes downstream and covers the distance between two ports in 4 h while it covers the same distance upstream in 5 h. If the speed of the stream is 2 km/h, then the speed of the streamer in still water is
 (a) 20 km/h (b) 19 km/h (c) 18 km/h (d) 19.5 km/h
- 20.** The solution of the system of linear equations $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$ is
 (a) $x = 3, y = 2$ (b) $x = 2, y = -3$
 (c) $x = 2, y = 3$ (d) None of these
- 21.** If $\sqrt{2}x - \sqrt{3}y = 0$ and $\sqrt{7}x + \sqrt{2}y = 0$, then find the value of $x + y$.
 (a) 1 (b) 2 (c) 3 (d) 0
- 22.** A man starts his job with a certain monthly salary and earns a fixed increment every year. If his salary was ₹ 1500 after 4 yr of services and ₹ 1800 after 10 yr of service. What was his starting salary?
 (a) ₹ 1300 (b) ₹ 1200 (c) ₹ 50 (d) ₹ 1100
- 23.** If $\frac{44}{x+y} + \frac{30}{x-y} = 10, \frac{55}{x+y} + \frac{40}{x-y} = 13$, then find the value of x and y .
 (a) $x = 2, y = 8$ (b) $x = 8, y = 3$
 (c) $x = 8, y = 2$ (d) $x = 3, y = 8$
- 24.** Consider the following sets of equations
 I. $2x - y = 0$ and $6x - 3y = 0$
 II. $3x - 4y = 0$ and $12x - 20y = 0$ Then,
 (a) Both sets I and II possess unique solution
 (b) Set I possesses unique solution and set II has infinitely many solutions
 (c) Set I possesses infinitely many solutions and set II possess unique solution
 (d) None of the sets I and II possesses a unique solution
- 25.** A fraction becomes $\frac{4}{5}$, if 1 is added to both numerator and denominator. If however 5 is subtracted from both numerator and denominator the fraction becomes $\frac{1}{2}$. Then, the fractions is
 (a) $\frac{7}{9}$ (b) $\frac{9}{7}$ (c) $\frac{3}{5}$ (d) $\frac{4}{3}$
- 26.** For what value of k , the following system of equations $3x + 4y = 6$ and $6x + 8y = k$ represents coincident lines?
 (a) 12 (b) 11 (c) 13 (d) 10
- 27.** The area of a rectangle remains the same if the length is increased by 7 m and the breadth is decreased by 3 m. The area remains unaffected if the length is decreased by 7 m and the breadth is increased by 5 m, then area of rectangle is
 (a) 280 m^2 (b) 320 m^2 (c) 420 m^2 (d) 400 m^2
- 28.** The value of x in the solution of the equation $2^{x+y} = 2^{x-y} = \sqrt{8}$ is
 (a) 0 (b) $\frac{3}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$
- 29.** A and B each have a certain number of mangoes. A says to B : "If you give 30 of your mangoes, I will have twice as many as left with you" B replies "If you give me 10, I will have thrice as many as left with you". How many mangoes did A has?
 (a) 41 (b) 62 (c) 34 (d) 32
- 30.** There are two examination rooms A and B. If 10 candidates are sent from room A to room B, the number of students in each room is the same. If 20 candidates are sent from B to A, the number of students in A is double the number of students in B. Then, number of students in room B is
 (a) 40 (b) 100 (c) 80 (d) 60

- 31.** A train started from a station with a certain number of passengers. At the first halt, $\frac{1}{3}$ rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started ?
 (a) 540 (b) 480 (c) 360 (d) 240
- 32.** The system of equations $x + 2y = 3$ and $3x + 6y = 9$ has
 (a) unique solutions (b) no solution
 (c) infinitely many solutions (d) finite number of solutions
- 33.** The sum of digits of a two-digit number is 8 and the difference between the number and that formed by reversing the digits is 18. What is the difference between the digits of the number?
 (a) 1 (b) 2 (c) 3 (d) 4
- 34.** If $(x, y) = (4, 1)$ is the solution of the pair of linear equations $mx + y = 2x + ny = 5$, then what is $m + n$ equal to?
 (a) -2 (b) -1 (c) 1 (d) 1
- 35.** If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then
 (a) $a=2, b=3$ (b) $a=-2, b=3$
 (c) $a=2, b=-3$ (d) $a=3, b=2$
- 36.** The sum of two numbers is 80. If the larger number exceeds four times the smaller by 5, what is the smaller number ?
 (a) 5 (b) 15 (c) 20 (d) 25
- 37.** If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, where $x, y \neq 0$ and $y \neq 0$, then what is the value of $x + y$?
 (a) 2 (b) 3 (c) 4 (d) 8
- 38.** Under what condition do the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution?
 (a) $k = 3$ (b) $k \neq 3$ (c) $k = 0$ (d) $k \neq 0$
- 39.** Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear. What can be the number of solutions (if they do exist)?
 (a) One or infinite (b) Only one
 (c) Exactly two (d) Exactly three
- 40.** A number consists of two digits, whose sum is 10. If 18 is subtracted from the number, digits of the number are reversed. What is the product?
 (a) 15 (b) 18
 (c) 24 (d) 32
- 41.** What is the value of k for which the system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution?
 (a) $-\frac{3}{14}$ (b) $-\frac{14}{3}$ (c) $\frac{1}{10}$ (d) 10
- 42.** What is the solution of the equation $x - y = 0.9$ and $11(x + y)^{-1} = 2$?
 (a) $x = 3.2$ and $y = 2.3$ (b) $x = 1$ and $y = 0.1$
 (c) $x = 2$ and $y = 1.1$ (d) $x = 1.2$ and $y = 0.3$
- 43.** Pooja started her job with certain monthly salary and gets a fixed increment every year. If her salary was ₹ 4200 after 3 yr and ₹ 6800 after 8 yr of service, then what are her initial salary and the annual increment, respectively ?
 (a) ₹ 2640, ₹ 320 (b) ₹ 2460, ₹ 320
 (c) ₹ 2460, ₹ 520 (d) ₹ 2640, ₹ 520
- 44.** A person bought 5 tickets from the station P to a station Q and 10 tickets from the station P to a station R . He paid ₹ 350. If the sum of fare of a ticket from P to Q and a ticket from P to R is ₹ 42, then what is the fare from P to Q ?
 (a) ₹ 12 (b) ₹ 14
 (c) ₹ 16 (d) ₹ 18
- 45.** The Community Relief fund receives a large donation of \$ 2800. The foundation agrees to spend the money on \$ 20 school bags, \$ 25 sweaters, \$ 5 books. They want to buy 200 items and send them to schools in earthquake-hit areas. They must order as many books as school bages and sweaters combined.
 How many of each item should they order?
 (a) (40, 60, 100) (b) (20, 30, 80)
 (c) (50, 100, 60) (d) 40, 80, 25)
- Directions** (Q. Nos. 46-47) *Some money is divided among Rajesh, Sonal and Chetan in such a way that 2 times share of Rajesh, 3 times share of Sonal and 5 times share of Chetan are all equal.*
Now, answer the following questions based on above information.
- 46.** If the sum of 6 times the share with Rajesh and 6 times the share with Sonal is ₹ 150, then find the share of Chetan?
 (a) ₹ 5 (b) ₹ 6 (c) ₹ 10 (d) ₹ 12
- 47.** If the sum of shares of Rajesh, Chetan and Sonal is ₹ 155, then find the share of Rajesh.
 (a) ₹ 75 (b) ₹ 35 (c) ₹ 50 (d) ₹ 25

PREVIOUS YEARS' QUESTIONS

- 48.** The graphs of $ax + by = c$, $dx + ey = f$ will be
 I. Parallel, if the system has no solution.
 II. Coincident, if the system has finite number of solutions.
 III. Intersecting, if the system has only one solutions
 Which of the statement(s) given above is/are correct? **2012 I**
 (a) Both I and II (b) Both II and III
 (c) Both I and III (d) All of these
- 49.** The sum of two numbers is 20 and their product is 75. What is the sum of their reciprocals? **2012 I**
 (a) $\frac{1}{15}$ (b) $\frac{1}{5}$ (c) $\frac{4}{15}$ (d) $\frac{7}{15}$
- 50.** If $3^{x+y} = 81$ and $81^{x-y} = 3$, then what is the value of x ? **2012 I**
 (a) $\frac{17}{16}$ (b) $\frac{17}{8}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$
- 51.** If $\frac{a}{b} - \frac{b}{a} = \frac{x}{y}$ and $\frac{a}{b} + \frac{b}{a} = x - y$, then what is the value of x ? **2013 I**
 (a) $\frac{a+b}{a}$ (b) $\frac{a+b}{b}$ (c) $\frac{a-b}{a}$ (d) None of these
- 52.** The system of equations $3x + y - 4 = 0$ and $6x + 2y - 8 = 0$ has **2013 I**
 (a) a unique solution $x = 1$, $y = 1$
 (b) a unique solution $x = 0$, $y = 4$
 (c) no solution
 (d) infinite solutions
- 53.** The sum of two numbers is 7 and the sum of their squares is 25. The product of the two numbers is **2013 I**
 (a) 6 (b) 10 (c) 12 (d) 15

Directions (Q. Nos. 54-55) A number consists of two digits whose sum is 10. If the digits of the number are reversed, then the number is decreased by 36.

- 54.** Which of the following is/are correct?
 I. The number is divisible by a composite number.
 II. The number is a multiple of a prime number.
 Select the correct answer using the codes given below **2013 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 55.** What is the products of the two digits? **2013 I**
 (a) 21 (b) 24 (c) 36 (d) 42
- 56.** Ten chairs and six tables together cost ₹ 6200, three chairs and two tables together cost ₹ 1900. The cost of 4 chairs and 5 tables is **2013 I**
 (a) ₹ 3000 (b) ₹ 3300 (c) ₹ 3500 (d) ₹ 3800

- 57.** A bus starts with some passengers. At the first stop, one-fifth of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now is 70. The number of passengers with which the bus started was **2013 II**
 (a) 40 (b) 50 (c) 60 (d) 70
- 58.** If $x + y - 7 = 0$ and $3x + y - 13 = 0$, then what is $4x^2 + y^2 + 4xy$ equal to? **2013 II**
 (a) 75 (b) 85 (c) 91 (d) 100
- 59.** If $\frac{x}{2} + \frac{y}{3} = 4$ and $\frac{2}{x} + \frac{3}{y} = 1$, then what is $x + y$ equal to? **2013 II**
 (a) 11 (b) 10 (c) 9 (d) 8
- 60.** If $x + y = 5$, $y + z = 10$ and $z + x = 15$, then which one of the following is correct? **2014 I**
 (a) $z > x > y$ (b) $z > y > x$ (c) $x > y > z$ (d) $x > z > y$
- 61.** The present age of Ravi's father is 4 times Ravi's present age. 5 yr back, Ravi's father was seven times as old as Ravi was at that time. What is the present age of Ravi's father? **2014 I**
 (a) 84 yr (b) 70 yr (c) 40 yr (d) 35 yr
- 62.** The sum of two positive number x and y is 2.5 times their difference. If the product of numbers is 84, then what is the sum of those two numbers? **2014 I**
 (a) 26 (b) 24 (c) 22 (d) 20
- 63.** Two chairs and one table cost ₹ 700 and one chair and Two tables cost ₹ 800. If cost m tables and m chairs is ₹ 30000, then what is m equal to? **2014 I**
 (a) 60 (b) 55 (c) 50 (d) 45
- 64.** A certain number of two digits is three times the sum of its digits. If 45 is added to the number, then the digits will be reversed. What is the sum of the squares of the two digits of the number? **2014 II**
 (a) 41 (b) 45 (c) 53 (d) 64
- 65.** A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**
 (a) 12 (b) 15 (c) 25 (d) 32
- 66.** A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was $\frac{3}{4}$ full. Oil of how many bottles can the tin contain? (All bottles are of equal volume) **2015 II**
 (a) 35 (b) 40 (c) 45 (d) 50

- 67.** A number consists of two digits, whose sum is 7. If the digits are reversed, the number is increased by 27. The product of digits of the number is **2015 II**
 (a) 6 (b) 8 (c) 10 (d) 12
- 68.** If $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = n$, then what is $\frac{x}{y}$ equal to? **2016 I**
 (a) $\frac{np + mq}{mp + nq}$ (b) $\frac{np + mq}{mp - nq}$
 (c) $\frac{np - mq}{mp - nq}$ (d) $\frac{np - mq}{mp + nq}$
- 69.** The value of k , for which the system of equations $3x - ky - 20 = 0$ and $6x - 10y + 40 = 0$ has no solution, is **2016 I**
 (a) 10 (b) 6 (c) 5 (d) 3
- 70.** There are three brothers. The sums of ages of two of them at a time are 4 yr, 6 yr and 8 yr. The age difference between the eldest and the youngest is **2016 I**
 (a) 3 yr (b) 4 yr (c) 5 yr (d) 6 yr
- 71.** The annual incomes of two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : 3. If each of them saves ₹ 2000 per year, then what is the difference in their annual incomes? **2016 I**
 (a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500
- 72.** Let a two digits number be k times the sum of its digits. If the number formed by interchanging the digits is m times the sum of the digits, then the value of m is **2016 I**
 (a) $9 - k$ (b) $10 - k$ (c) $11 - k$ (d) $k - 1$

ANSWERS

1	a	2	d	3	b	4	d	5	a	6	c	7	c	8	b	9	b	10	d
11	b	12	c	13	a	14	b	15	c	16	b	17	a	18	a	19	c	20	c
21	d	22	a	23	b	24	c	25	a	26	a	27	c	28	b	29	c	30	c
31	d	32	c	33	b	34	a	35	d	36	b	37	c	38	b	39	b	40	c
41	d	42	a	43	d	44	b	45	a	46	b	47	a	48	c	49	c	50	b
51	d	52	d	53	c	54	b	55	a	56	a	57	b	58	d	59	b	60	a
61	c	62	d	63	a	64	c	65	a	66	b	67	c	68	c	69	c	70	b
71	a	72	c																

HINTS AND SOLUTIONS

- 1. (a)** We have, $25x - 19 - [3 - \{4x - 5\}]$
 $= 3x - (6x - 5)$
 $\Rightarrow 25x - 19 - [3 - 4x + 5]$
 $= 3x - 6x + 5$
 $\Rightarrow 25x - 19 + 4x - 8 = -3x + 5$
 $\Rightarrow 29x + 3x = 5 + 27$
 $\Rightarrow 32x = 32 \Rightarrow x = \frac{32}{32} = 1 \Rightarrow x = 1$
 Hence, the value of x is 1.
- 2. (d)** Let the number be x .
 According to the question,
 $\frac{3x}{4} - x = -130 \Rightarrow \frac{3x - 4x}{4} = -130$
 $\Rightarrow \frac{-x}{4} = -130 \Rightarrow x = 520$
 Hence, the number is 520.

- 3. (b)** Let numbers be x and y , respectively.
 Then by given condition,
 $x + y = 21 \quad \dots(i)$
 $x - y = 11 \quad \dots(ii)$
 On adding Eqs. (i) and (ii), we get
 $2x = 32$
 $x = \frac{32}{2} = 16$
 On putting $y = 16$ in Eq. (i), we get
 $x + 16 = 21$
 $\therefore x = 5$
 Hence, the greatest number is 16.
- 4. (d)** Put $x = 2$ and $y = 1$ in each equation
 I. $2x + 5y = 9 \Rightarrow 2(2) + 5(1) = 9$
 $9 = 9$, it is true.
 II. $5x + 3y = 14$
 $\Rightarrow 5(2) + 3(1) = 14$

- $\Rightarrow 13 = 14$, it is false.
- III. $2x + 3y = 7 \Rightarrow 2(2) + 3(1) = 7$
 $7 = 7$, it is true.
- IV. $2x - 3y = 1 \Rightarrow 2(2) - 3(1) = 1$
 $1 = 1$, it is true.
 So, $x = 2$ and $y = 1$
 is a solution of I, III and IV.
- 5. (a)** Given equations are,
 $x + 2y = 16 \quad \dots(i)$
 and $x + \frac{y}{2} = \frac{7}{10} \quad \dots(ii)$
 On multiplying Eq. (i) by 10 and Eq. (ii) by 10, we get
 or $10x + 20y = 160 \quad \dots(iii)$
 and $10x + 5y = 7 \quad \dots(iv)$

On subtracting, Eq. (iv) from Eq. (iii), we get

$$(10x + 20y) - (10x + 5y) = 16 - 7$$

$$\Rightarrow 15y = 9 \Rightarrow y = \frac{9}{15} \Rightarrow y = \frac{3}{5}$$

On putting the value of y in Eq. (i), we get

$$x + 2y = \frac{8}{5}, \quad x + \frac{6}{5} = \frac{8}{5}$$

$$\Rightarrow x = \frac{8}{5} - \frac{6}{5} = \frac{2}{5}$$

6. (c) A pair of linear equations in two variables has a unique solution if their graphs intersect in one point.

7. (c) Here, $\frac{a_1}{a_2} = \frac{6}{9} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{5}{15/2} = \frac{2}{3}$

and $\frac{c_1}{c_2} = \frac{11}{21} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

So, the system has no solution.

8. (b) Let the numbers be x and $2490 - x$.
6.5% of $x = \frac{65}{100} \times x = \frac{13x}{200}$

$$8.5\% \text{ of } (2490 - x) = \frac{85}{100} (2490 - x)$$

$$= \frac{17}{200} (2490 - x)$$

By given condition, $\frac{13x}{200} = \frac{17(2490 - x)}{200}$

$$\Rightarrow 13x = 17(2490 - x)$$

$$\Rightarrow 13x + 17x = 42330$$

$$\Rightarrow x = \frac{42330}{30} = 1411$$

$$\therefore \text{Second number} = (2490 - x)$$

$$= 2490 - 1411 = 1079$$

9. (b) As this is the case of unique solution $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, so the graph of the equations

will intersect at one point.

10. (d) Given equations are,

$$4x + 6y = 11 \text{ and } 2x + ky = 7$$

Here, $a_1 = 4, b_1 = 6, c_1 = 11$

and $a_2 = 2, b_2 = k, c_2 = 7$

Here, $\frac{a_1}{a_2} = \frac{4}{2}, \frac{b_1}{b_2} = \frac{6}{k} \text{ and } \frac{c_1}{c_2} = \frac{11}{7}$

Since, system is inconsistent, so

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{4}{2} = \frac{6}{k}$$

$$\Rightarrow 4k = 12 \Rightarrow k = 3$$

Hence, the value of k is 3.

11. (b) Given equations are,

$$2x - ky = 4 \text{ and } 3x + 2y = 6$$

Here, $a_1 = 2, b_1 = -k, c_1 = 4$

and $a_2 = 3, b_2 = 2, c_2 = 6$

Now, $\frac{a_1}{a_2} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{-k}{2} \text{ and } \frac{c_1}{c_2} = \frac{4}{6}$

Since, the system has infinitely many solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}, \quad \frac{2}{3} = \frac{-k}{2} = \frac{2}{3}$$

$$\Rightarrow -3k = 4 \Rightarrow k = \frac{-4}{3}$$

12. (c) Given equations are,

$$\alpha x + 3y = \alpha - 3 \text{ and } 12x + \alpha y = \alpha$$

Here, $a_1 = \alpha, b_1 = 3, c_1 = \alpha - 3$

$$a_2 = 12, b_2 = \alpha, c_2 = \alpha$$

Since, system has unique solution,

So, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \frac{\alpha}{12} \neq \frac{3}{\alpha}$

$$\Rightarrow \alpha^2 \neq 36 \Rightarrow \alpha \neq \pm 6$$

13. (a) I. $2x + 3y = 425, 3x + 2y = 350$

Solving both the equations, we get
 $x = 40$ and $y = 115$.

II. When $k = -2$

$$\frac{a_1}{a_2} = \frac{1}{3}, \frac{b_1}{b_2} = \frac{2}{6} = \frac{1}{3}, \frac{c_1}{c_2} = \frac{-2}{5}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Hence, the given equations have no solution.

III. $(2, 5)$ is not a solution of
 $2x + 5y = 13$ as

$$2(2) + 5(5) = 4 + 25 = 29 \neq 13.$$

Hence, only I is correct.

14. (b) Let cost of one horse and one cow be ₹ x and ₹ y , respectively.

Condition I $x + 2y = 680$... (i)

Condition II $x = y + 80$

$$\Rightarrow x - y = 80$$
 ... (ii)

On subtracting Eq. (ii) from Eq. (i), we get

$$(x + 2y) - (x - y) = 680 - 80$$

$$\Rightarrow 3y = 600$$

$$\therefore y = 200$$

On putting $y = 200$ in Eq. (i), we get

$$x + 2 \times 200 = 680$$

$$\Rightarrow x + 400 = 680 \Rightarrow x = 680 - 400$$

$$\therefore x = 280$$

Hence, the cost of one horse is ₹ 280.

15. (c) Let number of 10 paise coins be x and number of 50 paise coins be y .

According to the questions,

Then, $x + y = 17$... (i)

and $10x + 50y = 450$... (ii)

From Eq. (ii), $x + 5y = 45$... (iii)

On subtracting Eq. (i) from Eq. (iii), we get

$$(x + 5y) - (x + y) = 45 - 17$$

$$\Rightarrow 4y = 28 \Rightarrow y = \frac{28}{4} = 7$$

$\therefore$ Number of 10 paise coins

$$= x = 17 - y = 17 - 7 = 10$$

16. (b) Given,

$$\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$$

(I) (II) (III)

Taking Ist and IInd terms,

$$5(3x - y + 1) = 3(2x + y + 2)$$

$$\Rightarrow 9x - 5y = 1$$
 ... (i)

Taking IInd and IIIrd terms

$$6(2x + y + 2) = 5(3x + 2y + 1)$$

$$\Rightarrow 3x + 4y = 7$$
 ... (ii)

On solving Eqs. (i) and (ii), we get $y = 1$ and $x = 1$

17. (a) Given, $3x - 5y = -10$

At Y-axis, $x = 0$

$$\Rightarrow 0 - 5y = -10 \Rightarrow y = \frac{10}{5} = 2$$

So, the point on Y-axis is $(0, 2)$.

18. (a) Given, $px + q = 0$ and $rx + s = 0$

$$\Rightarrow x = \frac{-q}{p} \text{ and } x = \frac{-s}{r}$$

So, $\frac{-q}{p} = \frac{-s}{r} \Rightarrow ps = qr$

19. (c) Let the speed of the streamer in still water = x km/h

then speed of streamer downstream

$$= (x + 2) \text{ km/h}$$

speed of streamer upstream

$$= (x - 2) \text{ km/h}$$

Distance travelled by streamer

downstream in 4h

$$= 4(x + 2) \text{ km}$$

$$[\because \text{distance} = \text{time} \times \text{speed}]$$

and distance travelled by streamer upstream in 5 h = $5(x - 2)$

So, $4(x + 2) = 5(x - 2)$

$$\Rightarrow 4x + 8 = 5x - 10$$

Hence, $x = 18$ km/h is the speed of streamer in still water.

20. (c) Given, $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$

On multiplying above equation by 10, we get

$$4x + 3y = 17$$
 ... (i)

and $7x - 2y = 8$... (ii)

On multiplying Eq. (i) by 2 and Eq. (ii) by 3, we get